

- 500 000 years ago, the European house mouse, *Mus musculus*, evolved into two subspecies, *Mus musculus domesticus* and *Mus musculus musculus*.

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- (b) Today, *M. m. domesticus* populations are separated from *M. m. musculus* populations by a large hybrid zone. The hybrid zone formed 5000–1000 years ago when populations of the two subspecies overlapped and interbreeding occurred between the two subspecies, resulting in hybrids.

Researchers investigated the populations in the hybrid zone.

It was observed that:

- hybrid mice were infected by more intestinal worms than *M. m. domesticus* and *M. m. musculus*
- *M. m. domesticus* and *M. m. musculus* individuals frequently mate together
- hybrid male mice had a very low fertility score based on testis weight and total sperm production, whereas *M. m. domesticus* and *M. m. musculus* males had a very high fertility score
- some female hybrids were sterile
- crosses between a fertile female hybrid and a male from either subspecies produced a very low number of offspring.

M. m. domesticus and *M. m. musculus* usually have the same diploid number ($2n = 40$). Some individuals of *M. m. domesticus* have a different diploid number ($2n = 34$).

Discuss the extent to which pre-zygotic and post-zygotic isolating mechanisms maintain *M. m. domesticus* and *M. m. musculus* as two separate subspecies within the hybrid zone.

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